
Exercise 2.3

Rules for Finding the Characteristic

(i) For a number *greater* than 1:

$$\text{Characteristic} = \text{number of digits to the left of the decimal point} - 1$$

(ii) For a number *less* than 1:

$$\text{Characteristic} = -(\text{number of zeros between the decimal point and the first non zero digit} + 1)$$

Note: When the characteristic is negative, we write it with *bar*.

Logarithm of a Number

$$\text{Logarithm} = \text{Characteristic} + \text{Mantissa}$$

1. Find the characteristic of the following numbers:

(i) 5287

$$\begin{aligned}\text{Characteristic} &= 4 - 1 \\ &= 3\end{aligned}$$

(ii) 59.28

$$\begin{aligned}\text{Characteristic} &= 2 - 1 \\ &= 1\end{aligned}$$

(iii) 0.0567

$$\begin{aligned}\text{Characteristic} &= -(1 + 1) \\ &= \bar{2}\end{aligned}$$

(iv) 234.7

$$\begin{aligned}\text{Characteristic} &= 3 - 1 \\ &= 2\end{aligned}$$

(v) 0.000049

$$\begin{aligned}\text{Characteristic} &= -(4 + 1) \\ &= \bar{5}\end{aligned}$$

(vi) 145000

$$\begin{aligned}\text{Characteristic} &= 6 - 1 \\ &= 5\end{aligned}$$

2. Find the logarithm of the following numbers:

(i) 43

$$\begin{aligned}\text{Characteristic} &= 2 - 1 = 1 \\ \text{Mantissa} &= 0.6335 \quad (\text{Table Value} = 6335) \\ \log 43 &= 1 + 0.6335 \\ \log 43 &= 1.6335\end{aligned}$$

(ii) 579

$$\begin{aligned}\text{Characteristic} &= 3 - 1 = 2 \\ \text{Mantissa} &= 0.7627 \quad (\text{Table Value} = 7627) \\ \log 579 &= 2 + 0.7627 \\ \log 579 &= 2.7627\end{aligned}$$

(iii) 1.982

$$\begin{aligned}\text{Characteristic} &= 1 - 1 = 0 \\ \text{Mantissa} &= 0.2971 \quad (\text{Table Value} = 2967 + 4 = 2971)\end{aligned}$$

$$\log 1.982 = 0 + 0.2971$$

$$\log 1.982 = 0.2971$$

(iv) 0.0876

$$\text{Characteristic} = -(1 + 1) = -2$$

$$\text{Mantissa} = 0.9425 \quad (\text{Table Value} = 9425)$$

$$\log 0.0876 = -2 + 0.9425$$

$$\log 0.0876 = -1.0575$$

(v) 0.047

$$\text{Characteristic} = -(1 + 1) = -2$$

$$\text{Mantissa} = 0.6721 \quad (\text{Table Value} = 6721)$$

$$\log 0.047 = -2 + 0.6721$$

$$\log 0.047 = -1.3279$$

(vi) 0.000354

$$\text{Characteristic} = -(3 + 1) = -4$$

$$\text{Mantissa} = 0.5490 \quad (\text{Table Value} = 5490)$$

$$\log 0.000354 = -4 + 0.5490$$

$$\log 0.000354 = -3.4510$$

Note: The place between the first non-zero digit from left and its next digit is called reference position. For example, in 1332, the reference position is between 1 and 3 (1,332).

3. If $\log 3.177 = 0.5019$, then find:

Since $\log 3.177 = 0.5019$. So, *Characteristic* = 0 and *Mantissa* = 0.5019

(i) $\log 3177$

$$\text{Characteristic} = 4 - 1 = 3$$

$$\text{Mantissa} = 0.5019$$

$$\log 3177 = 3 + 0.5019$$

$$\log 3177 = 3.5019$$

(ii) $\log 31.77$

$$\text{Characteristic} = 2 - 1 = 1$$

$$\text{Mantissa} = 0.5019$$

$$\log 3177 = 1 + 0.5019$$

$$\log 3177 = 1.5019$$

(iii) $\log 0.03177$

$$\text{Characteristic} = -(1 + 1) = -2$$

$$\text{Mantissa} = 0.5019$$

$$\log 0.03177 = -2 + 0.5019$$

$$\log 0.03177 = -1.4981$$

4. Find the value of x :

(i) $\log x = 0.0065$

Since $\log x = 0.0065$. So, *Characteristic* = 0 and *Mantissa* = 0.0065

$$\text{Table value of } 0.0065 = 1014 + 1 = 1,015$$

So

$$x = \text{Anti log } 0.0065$$

$$x = 1.015$$

Since *characteristic* = 0, therefore decimal point will be at reference position.

(ii) $\log x = 1.192$

Since $\log x = 1.192$. So, *Characteristic* = 1 and *Mantissa* = 0.192

Table value of 0.192 = 1,556

So

$$x = \text{Anti log } 1.192$$

$$x = 15.56$$

Since *characteristic* = 1, therefore decimal point will be after 1 digits right from the reference position.

(iii) $\log x = -3.434$

$$\log x = -3.434$$

$$\log x = -4 + 4 - 3.434$$

$$\log x = -4 + 0.566$$

$$\log x = \bar{4}.566$$

Since $\log x = \bar{4}.566$. So, *Characteristic* = $\bar{4}$ and *Mantissa* = 0.566

Table value of 0.566 = 3,681

So

$$x = \text{Anti log } \bar{4}.566$$

$$x = 0.0003681$$

Since *characteristic* = $\bar{4}$, therefore decimal point will be before 4 digits left from the reference position.

(iv) $\log x = -1.5726$

$$\log x = -1.5726$$

$$\log x = -2 + 2 - 1.5726$$

$$\log x = -2 + 0.4274$$

$$\log x = \bar{2}.4274$$

Since $\log x = \bar{2}.4274$. So, *Characteristic* = $\bar{2}$ and *Mantissa* = 0.4274

Table value of 0.4274 = 2673 + 2 = 2,675

So

$$x = \text{Anti log } \bar{2}.4274$$

$$x = 0.02675$$

Since *characteristic* = $\bar{2}$, therefore decimal point will be before 2 digits left from the reference position.

(v) $\log x = 4.3561$

Since $\log x = 4.3561$. So, *Characteristic* = 4 and *Mantissa* = 0.3561

Table value of 0.3561 = 2270 + 1 = 2,271

So

$$x = \text{Anti log } 4.3561$$

$$x = 22710$$

Since *characteristic* = 4, therefore decimal point will be after 4 digits right from the reference position.

(vi) $\log x = -2.0184$

$$\log x = -2.0184$$

$$\log x = -3 + 3 - 2.0184$$

$$\log x = -3 + 0.9816$$

$$\log x = \bar{3}.9816$$

Since $\log x = \bar{3}.9816$. So, *Characteristic* = $\bar{3}$ and *Mantissa* = 0.9816

Table value of 0.9816 = 9572 + 13 = 9,585

So

$$x = \text{Anti log } \bar{3}.9816$$

$$x = 0.009585$$

Since *characteristic* = $\bar{3}$, therefore decimal point will be before 3 digits left from the reference position.